Chapter 5 Chemical reactions

EXTRA WORKSHEET 1: More balancing chemical equations

SCIENCE UNDERSTANDING **LOGICAL/MATHEMATICAL**

Name:_____Class:_____Date: _____

Balanced chemical equations show the atoms and molecules that take part in a chemical reaction. These equations take into account the Law of Conservation of Mass by ensuring that there is the same number of each atom in the products as there is in the reactants.

Equations are balanced by adding coefficients (numbers) to the chemical formulas in the equation. The coefficient multiplies every atom in the molecule. For example, putting a coefficient of three in front of the chemical formula for water ($3H_2O$) gives a total of 6 hydrogen atoms and 3 oxygen atoms.

Method 1: Using integer coefficients

Many chemical equations can be balanced by simply balancing the number of each type of atom in the reactants and products one by one. Take for example the combustion of methane, which has the unbalanced chemical equation:

CH_4	+	O_2	→	CO_2	+	H_2O
methane	+	oxygen	→	carbon dioxide	+	water

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Step 1: Balance the carbon atoms (C). There is one carbon in the reactants and one in the products, so no change is necessary.

CH_4	+	O_2	→	CO_2	+	H_2O
methane	+	oxygen	→	carbon dioxide	+	water

Step 2: Balance the hydrogen atoms (H). There are four hydrogen atoms in the reactants and two in the products so the water molecule is multiplied by two.

CH_4	+	O_2	→	CO_2	+	**2**H_2O
methane	+	oxygen	→	carbon dioxide	+	water

Step 3: Balance the oxygen atoms (O). There are two oxygen atoms in the reactants and four oxygen atoms in the products (two from the carbon dioxide and two more from the two water molecules). Therefore, the oxygen molecule in the reactants is multiplied by two.

CH_4	+	**2**O_2	→	CO_2	+	$2H_2O$
methane	+	oxygen	→	carbon dioxide	+	water

Step 4: Check that the equation is balanced by looking at the number of each atom on either side of the equation.

Reactants: 1 C, 4 H, 4 O

Products: 1 C, 4 H, 4 O

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Method 2: Using half-integer coefficients

One of the challenges when balancing chemical equations is that not all equations can be balanced using the same method. Therefore, scientists develop different strategies for dealing with different equations.

One useful strategy is to first balance the equation using half-integer coefficients such as 0.5, 1.5, 2.5 and so on. However, it doesn't make sense to have half a molecule so the whole equation is then multiplied by a factor of two.

This technique comes in handy when you find that there is an odd number of an atom of one side of the equation and an even number on the other side of the equation. Take for example, the combustion of ethane, which has the unbalanced chemical equation:

C_2H_6	+	O_2	→	CO_2	+	H_2O
ethane	+	oxygen	→	carbon dioxide	+	water

Step 1: Balance the carbons (C) by multiplying the carbon dioxide by two.

C_2H_6	+	O_2	→	**2**CO_2	+	H_2O
ethane	+	oxygen	→	carbon dioxide	+	water

Step 2: Balance the hydrogens (H) by multiplying the water by three.

C_2H_6	+	O_2	→	$2CO_2$	+	**3**H_2O
ethane	+	oxygen	→	carbon dioxide	+	water

Step 3: Balance the oxygens (O). There are 2 oxygens in the reactants and 7 oxygens in the products. There is no way we can balance the oxygens by multiplying any single molecule by an integer. Therefore we multiply the oxygen molecule by 3.5 to give 7 oxygen atoms in the reactants and 7 in the products.

C_2H_6	+	**3.5**O_2	→	$2CO_2$	+	$3H_2O$
ethane	+	oxygen	→	carbon dioxide	+	water

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Step 4: Multiply all molecules by 2 because it doesn't make sense to have three and a half oxygen molecules.

$2C_2H_6$	+	**7**O_2	→	$4CO_2$	+	$6H_2O$
ethane	+	oxygen	→	carbon dioxide	+	water

Step 5: Check that the equation is balanced.

Reactants: 4 C, 12 H, 14 O

Products: 4 C, 12 H, 14 O

Questions

1 **Use** the method of integer coefficients outlined above to balance the following chemical equations.

a

HNO_3	+	MgO	→	$Mg(NO_3)_2$	+	H_2O
nitric acid	+	magnesium oxide	→	magnesium nitrate	+	water

b

Mg	+	HCl	→	$MgCl_2$	+	H_2
magnesium	+	hydrochloric acid	→	magnesium chloride	+	hydrogen

c

$CaCO_3$	+	HCl	→	CO_2	+	H_2O	+	$CaCl_2$
calcium carbonate	+	hydrochloric acid	→	carbon dioxide	+	water	+	calcium chloride

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d

$C_6H_{12}O_6$	+	O_2	→	CO_2	+	H_2O
glucose	+	oxygen	→	carbon dioxide	+	water

e

P_4	+	O_2	→	P_2O_5
phosphorus	+	oxygen	→	phosphorus oxide

f

C_3H_{12}	+	O_2	→	CO_2	+	H_2O
propane	+	oxygen	→	carbon dioxide	+	water

g

O_2	+	Zn	→	ZnO
oxygen	+	zinc	→	zinc oxide

h

HNO_3	+	BaO	→	$Ba(NO_3)_2$	+	H_2O
nitric acid	+	barium oxide	→	barium nitrate	+	water

i

Na_2CO_3	+	HNO_3	→	CO_2	+	H_2O	+	$NaNO_3$
sodium carbonate	+	nitric acid	→	carbon dioxide	+	water	+	sodium nitrate

j

NaOH	+	H_2SO_4	→	Na_2SO_4	+	H_2O
sodium hydroxide	+	sulfuric acid	→	sodium sulfate	+	water

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2 **Use** the method of half-integers to balance the following equations showing:

i the balanced equation with half-integer coefficients

ii multiply all coefficients by 2 to make a balanced equation with only integer coefficients

a **i**

Al	+	O_2	→	Al_2O_3
aluminium	+	oxygen	→	aluminium oxide

ii

Al	+	O_2	→	Al_2O_3
aluminium	+	oxygen	→	aluminium oxide

b **i**

C_4H_{10}	+	O_2	→	CO_2	+	H_2O
butane	+	oxygen	→	carbon dioxide	+	water

ii

C_4H_{10}	+	O_2	→	CO_2	+	H_2O
butane	+	oxygen	→	carbon dioxide	+	water

c **i**

$KClO_3$	→	KCl	+	O_2
potassium chlorate	→	potassium chloride	+	oxygen

ii

$KClO_3$	→	KCl	+	O_2
potassium chlorate	→	potassium chloride	+	oxygen

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d i

Fe	+	O_2	→	Fe_2O_3
iron	+	oxygen	→	iron(III) oxide

ii

Fe	+	O_2	→	Fe_2O_3
iron	+	oxygen	→	iron(III) oxide

e i

Fe	+	$CuCl_2$	→	$FeCl_3$	+	Cu
iron	+	copper(II) chloride	→	iron(III) chloride	+	copper

ii

Fe	+	$CuCl_2$	→	$FeCl_3$	+	Cu
iron	+	copper(II) chloride	→	iron(III) chloride	+	copper

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